Umed Ali Keerio

AI/ML Engineer — Machine Learning Systems — Model Deployment & Production AI

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PROFESSIONAL SUMMARY

AI/ML Engineer and Software Engineering graduate with 2+ years of experience designing, training, and deploying machine learning and deep learning models into production systems. Hands-on background spanning classical ML, computer vision, and NLP model development (CNN, ResNet, LSTM, T5, RoBERTa) alongside LLM-based and agentic systems, with end-to-end ownership from data preprocessing and feature engineering through model training, evaluation, and deployment. Comfortable working with containerized deployment pipelines (Docker, Kubernetes) and cloud-based environments, collaborating cross-functionally with software engineers to ship reliable, production-grade AI features.

EDUCATION

Virtual University of Pakistan — BSc Software Engineering Mar 2022 – May 2026

- Relevant Coursework: AI, Database Management, Data Structures & Algorithms, Computer Networks, Information Security, Software Engineering

PROFESSIONAL EXPERIENCE

AI & Agentic Systems Engineer — Devzz Tech Solutions, Islamabad Feb 2025 – Mar 2026

- Designed, trained, and deployed AI/ML models and pipelines into production systems, integrating LLM APIs (OpenAI, Claude, Gemini) with scalable backend services via secure REST APIs
- Built retrieval and vector-search components (FAISS, ChromaDB, Pinecone) indexing 10K+ document chunks with sub-second retrieval, applying standard feature-engineering and evaluation practices to optimize accuracy
- Collaborated with frontend and backend engineers to integrate AI capabilities into web applications using FastAPI and Flask, taking models from prototype through to production-ready deployment
- Conducted iterative testing and validation to improve model output quality, reducing hallucination/error rates through systematic prompt and context-window optimization
- Containerized and orchestrated AI microservices using Docker and Kubernetes, working within cloud-based deployment environments and streamlining CI/CD pipelines to maintain 99%+ service uptime

AI Developer — Pixel TechnoSol, Islamabad Aug 2023 – May 2024

- Delivered end-to-end AI/ML projects spanning data preprocessing, model training, and deployment across NLP, Computer Vision, and Generative AI domains
- Built and evaluated ML/DL models for client-facing applications, using LangChain, OpenAI API, and HuggingFace Transformers alongside standard supervised-learning workflows
- Integrated trained models with web applications using Python, FastAPI, and Flask, exposing inference endpoints consumed by frontend teams
- Used automation workflows (n8n) to accelerate testing and delivery of client-facing AI solutions by an estimated 40%

Machine Learning Intern — Technocolabs Softwares, Remote Nov 2022 – Feb 2023

- Executed 3+ ML tasks covering data preprocessing, feature engineering, and model evaluation across datasets with 50K+ records
- Automated data cleaning pipelines in Python (Pandas, NumPy), reducing preprocessing time by approximately 30% on assigned datasets

KEY PROJECTS

Pathfinder AI – Autonomous Browser Automation Agent | Groq LLM, Playwright, WebSockets, React

- Built an autonomous web automation agent: give it a goal in natural language and it inspects the page, reasons about the next best action (Groq, llama-3.3-70b-versatile), and executes it in the browser with Playwright, replacing brittle hardcoded scripts
- Streams step-by-step progress, screenshots, and live logs over WebSockets to a React dashboard, with human-in-the-loop controls to pause, inspect, resume, or interrupt a run
- Added pre-built recipes (web scraping, job search, form submission, price comparison), run history with duration/status/step metrics, and configurable API keys, models, headless mode, and step limits

Brain Tumor Classification & Radiology Report Generation | ResNet101, T5, TensorFlow

- Architected an end-to-end supervised learning pipeline: ResNet101 classifying 4 MRI tumor classes with 96%+ accuracy, paired with a T5 model generating structured radiology reports, reducing manual reporting effort by an estimated 60%

DocuMind – RAG-Based PDF Intelligence System | LangChain, FAISS, Flask, Docker

- Built a retrieval pipeline for semantic Q&A over 100+ page PDF documents using LangChain, FAISS, and OpenAI embeddings, with configurable chunking (512–1024 tokens) and Docker-based deployment

PolyGuard AI – Cross-Lingual Plagiarism & AI Detection | React.js, FastAPI, RoBERTa

- Implemented cross-lingual semantic similarity detection across 50+ languages using paraphrase-multilingual-MiniLM embeddings, achieving 92%+ precision on benchmark test sets
- Integrated a RoBERTa-based AI-content classifier (95%+ accuracy) and AST-based code clone detection covering Python, JavaScript, and C++

MediIntel AI – Explainable Medical Diagnosis System | React.js, Flask, SpaCy, SHAP

- Built an NLP pipeline using SpaCy to extract 20+ symptom entities from free-text narratives, with SHAP-based explainability surfacing top-5 contributing features per diagnosis
- Implemented JWT auth with role-based access control and an analytics dashboard visualizing patient health trends across 6+ metrics

MovieSent – Dual-Model Sentiment Analysis API | LSTM, Logistic Regression, Flask

- Built and compared a dual-model sentiment classifier (LSTM + Logistic Regression) achieving ~99% accuracy; deployed as a production-grade Flask API with JWT-based authentication

Rice Plant Disease Detection | CNN, Flask, Streamlit

- Trained a CNN classifier (94%+ accuracy) for 3-class plant disease detection from image data; deployed as a full-stack app with Flask auth, Streamlit UI, and side-by-side image comparison

TECHNICAL SKILLS

Languages: Python (Expert), JavaScript, C++, SQL, HTML/CSS
ML/DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, Hugging Face Transformers, OpenCV, NLTK
Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation & Validation, Fine-tuning (LoRA/QLoRA)
NLP & Computer Vision: Transformers, Sentence-Transformers, SpaCy, SHAP, ResNet, T5, BERT, RoBERTa
LLM & GenAI: LangChain, LlamaIndex, OpenAI API, Claude API, Gemini API, Groq, RAG Pipelines, Prompt Engineering, AI Agents, Model Context Protocol (MCP), Claude Code
Automation: Playwright, n8n, WebSockets
Vector Databases: FAISS, Pinecone, Weaviate, ChromaDB
Data & Analysis: Pandas, NumPy, SQL, MongoDB, PostgreSQL, SQLite, Redis
Deployment & Cloud: Docker, Kubernetes, cloud-based deployment environments, REST APIs, CI/CD Pipelines, GitHub Actions
Tools: Git, Linux

CERTIFICATIONS

- **NCA-GENL: NVIDIA-Certified Generative AI & LLMs Specialization** – NVIDIA / Udemy (2026)
- **Generative AI Application Developer Certificate (Top Performer)** – NCEAC-HEC Cohort 3 / ULEFUSA (2026)
- **Claude Code in Action** – Anthropic (2026)
- **Introduction to Subagents** – Anthropic (2026)
- **Introduction to Model Context Protocol** – Anthropic (2026)
- **Applied AI Foundations** – OpenAI (2026)
- **Machine Learning with Python** – freeCodeCamp (2026)
- **Fine-tune BERT for Text Classification** – Coursera (2023)